

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Sixth Semester B.E. (AIML / CCE) (Credit System) Degree Examinations**

May - June 2023

20AM622 / 20CC622 - WIRELESS SENSOR NETWORKS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.*

Unit – I		Marks	BT*	CO*	PO*
1. a)	Discuss any FIVE major characteristics of wireless sensor networks.	10	L*1	1	1
b)	Explain how Sparse Topology and Energy Management (STEM) protocol provides a solution for the idle listening problem. Distinguish STEM-B and STEM-T variants.	10	L2	2	1
2. a)	How one can estimate the energy saving when a WSN node switches between active and sleep modes? Explain with a neat time diagram.	10	L2	1	1
b)	With neat diagram of S-MAC principle, explain the three phases of listening period of Sensor-MAC protocol.	10	L2	2	1
3. a)	Which one of these is the best choice between Microcontrollers, FPGAs, ASICs and DSPs as a controller for a WSN node? Justify your answer.	10	L4	1	1
b)	Illustrate the data exchange between coordinator and devices in a star network with the beacons mode of IEEE 802.15.4.	10	L2	2	1
Unit – II					
4. a)	Outline how Low-energy adaptive clustering hierarchy (LEACH) routing algorithm is designed to collect and deliver data to the data sink in WSNs.	10	L2	3	1
b)	Briefly explain the threshold sensitive energy efficient sensor network protocol (TEEN).	10	L2	4	1
5. a)	With a suitable example illustrate the chain-based data gathering and aggregation scheme of Power-Efficient Gathering in Sensor information Systems.	10	L3	3	1
b)	Illustrate the rules required for a distributed estimation algorithm in data aggregation for WSN data.	10	L2	4	1
6. a)	Explain the energy savings approach adapted by directed diffusion protocol in order to extend the lifetime of the network.	10	L2	3	1
b)	Illustrate the problem of greedy forwarding in dead ends and the solution suggested by greedy perimeter stateless routing (GPSR) protocol.	10	L2	4	1
Unit – III					
7. a)	What are the minimum set of functions that a sensor operating systems (SOS) should embed? Explain.	10	L2	5	1
b)	Construct a NesC module and configuration that blinks a LED for execution in TinyOS.	10	L3	5	1
8. a)	Why traditional OS are not suitable for WSN? Discuss any FOUR major design issues with OS for WSN.	10	L2	5	1
b)	Write short notes on i) EYES OS ii) MagnetOS	10	L1	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
